Data Dictionary

NY zonal congestion and Moses South flow, 2023–2025 · Use Case hackathon kit

Item	Detail
Price source	NYISO real-time market zonal LBMP, 5-minute intervals, 11 internal zones (A–K)
Flow source	NYISO interface flow data for the Moses South interface, 5-minute intervals
Coverage	Jan 1 2023 – Dec 31 2025 (36 months)
Units	Prices and congestion cost in $/MWh. Flow in MW, averaged over the hour or day
Time	Local time; hours are “hour beginning” (hour 17 = 5:00–6:00 pm)

Sign convention

LBMP = energy + losses − congestion

- **Negative congestion cost = congestion raised the price in that zone.** The more negative, the heavier the congestion.
- **Positive congestion cost = congestion lowered the price in that zone**, typically where power is trapped, such as the North Country.
- Zero = no congestion effect in that hour.

How the files were built

1. 5-minute real-time prices and flows were downloaded from NYISO for every zone and for Moses South.
2. Hourly values (the raw files) are averages of the 5-minute values in each hour.
3. Daily, monthly and hour-of-day values are averages of the 5-minute data in each period.
4. Monthly minimum and maximum columns are single 5-minute values, so they show the true extremes.

Congestion_monthly_by_zone_2023_2025.csv

396 rows, one per zone per month (long format).

Column	Meaning
year, month, season	Date parts. Seasons: Winter (Dec–Feb), Spring (Mar–May), Summer (Jun–Aug), Fall (Sep–Nov)
zone	Zone column name, such as J_new_york_city (see zones_reference.csv)
avg_lbmp	Average total price, $/MWh
min_lbmp, max_lbmp	Lowest and highest 5-minute price in the month, $/MWh

Column	Meaning
avg_congestion_cost	Average congestion cost, $/MWh (negative = raised the price)
min_congestion_cost, max_congestion_cost	Lowest and highest 5-minute congestion cost in the month, $/MWh

Congestion_Daily_by_zone_2023_2025.csv

1,096 rows, one per day (wide format). Columns: date, day_of_week, is_weekend (true or false), then one column per zone with that day's average congestion cost in $/MWh: A_west, B_genesee, C_central, D_north, E_mohawk_valley, F_capital, G_hudson_valley, H_millwood, I_dunwoodie, J_new_york_city, K_long_island.

Congestion_Hourly_by_Month_2023_2025.csv

864 rows = 36 months × 24 hours. Columns: year, month, hour (0–23, hour beginning), then the same 11 zone columns holding the average congestion cost for that hour of the day across the month, in $/MWh.

Congestion_Raw_2023_2025.csv

26,276 rows, one per hour. Columns: hour_timestamp (YYYY-MM-DD HH:MM), then the 11 zone columns with that hour's average congestion cost in $/MWh. 28 hours are missing: the three spring-forward daylight-saving hours, which do not exist, and 25 hours absent from the source data.

Mosesouth_flow_2023_2025_raw.csv

26,296 rows, one per hour; 8 hours are missing.

Column	Meaning
formatted_timestamp	Date and hour, YYYY-MM-DDTHH:MM
date, year, month, day, hour	Date parts; hour is 0–23, hour beginning
Interface Name	Always MOSES SOUTH
avg_flow_mwh	Average power flow across the interface during the hour, in MW (equal to the MWh moved in that hour). Positive = flow southbound out of the North Country; negative = flow northbound

Mosesouth_flow_2023_2025_dailyavg.csv

1,096 rows, one per day. Columns: date, day_of_week, is_weekend (true or false), Interface Name (MOSES SOUTH) and avg_flow_mwh, the day's average flow in MW. Join it to the daily congestion file on date.

zones_reference.csv

One row per zone: zone_letter, zone_name, nyiso_code (the name in NYISO's raw files), column_name (as used in the other files), region, representative_city, latitude, longitude. Coordinates mark a representative city for mapping; they are not the zone's boundary or center.

Zone	Name	Region	Representative city
A	West	Upstate	Buffalo
B	Genesee	Upstate	Rochester
C	Central	Upstate	Syracuse
D	North	Upstate	Plattsburgh
E	Mohawk Valley	Upstate	Utica
F	Capital	Upstate	Albany
G	Hudson Valley	Lower Hudson Valley	Poughkeepsie
H	Millwood	Lower Hudson Valley	Mount Kisco
I	Dunwoodie	Lower Hudson Valley	Yonkers
J	New York City	New York City	Manhattan
K	Long Island	Long Island	Hauppauge

Limits of this data

- Real-time prices are volatile and react to many factors in the moment.
- Prices are per MWh only; zone electricity usage is not included, so totals in dollars cannot be calculated.
- The data shows where and when congestion occurred, not which transmission line or facility caused it.
- Moses South carries all power flowing south from the North Country, not only Smart Path Connect. A link between flow and congestion is a correlation, not proof of cause.